Eigenvalues & Eigenvectors — 80/20 Interview Notes

For interviews, you do **not** need to memorize every theorem in this chapter. The high-value chain is:

Linear transformation → special direction → eigenvector → scaling factor → eigenvalue → characteristic equation → eigenspace → diagonalization → ML applications

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1. The Core Idea

Suppose we have a matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Normally, multiplying a vector by Acan **change both its direction and magnitude**.

But some special vectors are different.

For example:

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then:

$$Ax = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The direction did **not change**.

Only the magnitude changed by 3.

That is the entire intuition behind eigenvectors and eigenvalues.

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2. Definition — Most Important

An **eigenvector** of a matrix $A$is a non-zero vector $x$such that:

$$\boxed{Ax = \lambda x}$$

where:

- A= matrix
- x= eigenvector

- λ= eigenvalue
- $x \neq 0$

In simple English

An eigenvector is a direction that a transformation does not rotate away from; it only stretches, shrinks, or reverses that direction.

The eigenvalue tells us **how much**.

Eigenvalue	Meaning
$\lambda > 1$	Stretch
$0 < \lambda < 1$	Shrink
$\lambda = 1$	No change
$\lambda = 0$	Collapses to zero
$\lambda < 0$	Reverses direction + scales

Interview answer

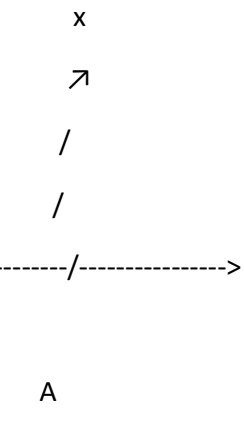
“An eigenvector is a non-zero vector whose direction remains unchanged under a linear transformation, while the eigenvalue specifies the scaling factor.”

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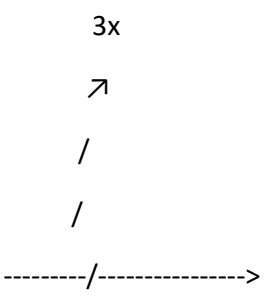
3. Diagram — Geometric Meaning

Think of a transformation acting on vectors:

Before transformation



After transformation



Same direction

3 × magnitude

Mathematically:

$$Ax = 3x$$

So $x$is an eigenvector and 3is its eigenvalue.

Important distinction

An eigenvector does **not necessarily stay exactly the same**.

It can become:

$$3x,\ 0.5x,\ -2x$$

The important thing is:

direction remains on the same line

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4. Example 1 — Find Eigenvectors by Inspection

Consider:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Try:

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then:

$$Ax_1 = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3x_1$$

Therefore:

$$\boxed{\lambda = 3}$$

and

$$\boxed{x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

is an eigenvector.

Now try:

$$x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Then:

$$Ax_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = x_2$$

Therefore:

$$\boxed{\lambda = 1}$$

and

$$\boxed{x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}}$$

is an eigenvector.

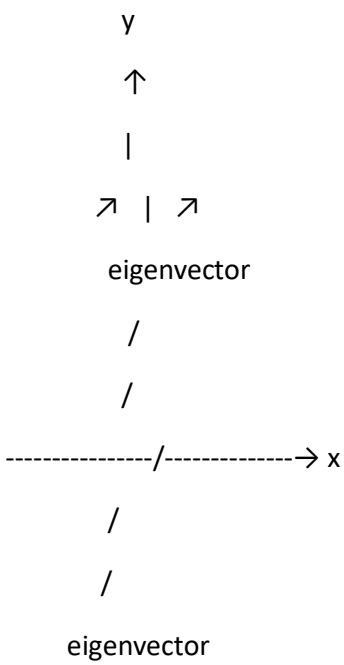
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5. The 80/20 Geometric Picture

For this matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

there are two special directions:



More specifically:

Direction 1

$$(1, 1)$$

gets stretched by 3.

Direction 2

$$(-, 11)$$

gets multiplied by 1.

So the transformation becomes extremely simple if we describe vectors using these two directions instead of the normal x, yaxes.

This leads directly to **diagonalization**.

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6. How Do We Find Eigenvalues?

This is one of the most important interview procedures.

Start with:

$$Ax = \lambda x$$

Move everything to one side:

$$Ax - \lambda x = 0$$

Since:

$$\lambda x = \lambda I x$$

we get:

$$(A - \lambda I)x = 0$$

For a non-zero solution $x$to exist:

$$A - \lambda I$$

must be singular.

Therefore:

$$\boxed{\det(A - \lambda I) = 0}$$

This is the **characteristic equation**.

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7. Characteristic Polynomial

Define:

$$\boxed{p(\lambda) = \det(A - \lambda I)}$$

This is called the **characteristic polynomial**.

Its roots are the eigenvalues.

Interview chain

Memorize this:

$$\boxed{Ax = \lambda x}$$

↓

$$\boxed{(A-\lambda I)x=0}$$

↓

Non-zero solution exists

↓

$$\boxed{\det(A-\lambda I)=0}$$

↓

Solve polynomial

↓

Eigenvalues

This is probably the single most important derivation in the chapter.

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8. Example 2 — Finding Eigenvalues

Given:

$$A=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$$

Calculate:

$$A-\lambda I=\begin{bmatrix}2-\lambda & 1\\ 1 & 2-\lambda\end{bmatrix}$$

Take determinant:

$$\det(A-\lambda I)=(2-\lambda)^2-1$$

Expand:

$$=4-4\lambda+\lambda^2-1$$
$$=\lambda^2-4\lambda+3$$

Set equal to zero:

$$\lambda^2 - 4\lambda + 3 = 0$$

Factor:

$$(\lambda - 3)(\lambda - 1) = 0$$

Therefore:

$$\boxed{\lambda_1 = 3,\ \lambda_2 = 1}$$

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9. Finding Eigenvectors

Once you know an eigenvalue, substitute it into:

$$\boxed{(A - \lambda I)x = 0}$$

This gives a system of linear equations.

Example — Eigenvector for $\lambda = 3$

$$A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

Solve:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We get:

$$-x_1 + x_2 = 0$$

Therefore:

$$x_1 = x_2$$

So:

$$x = \begin{bmatrix} t \\ t \end{bmatrix}$$

for any $t \neq 0$.

Choose $t = 1$:

$$\boxed{x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

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10. Important: Eigenvector Is Not Unique

If:

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is an eigenvector, then:

$$2x = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

is also an eigenvector.

And:

$$-5x = \begin{bmatrix} -5 \\ -5 \end{bmatrix}$$

is also an eigenvector.

Because:

$$A(cx) = cAx = c\lambda x = \lambda(cx)$$

Therefore, eigenvectors corresponding to an eigenvalue form a **subspace**.

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11. Eigenspace

The eigenspace corresponding to eigenvalue $\lambda$is:

$$\boxed{E_\lambda = \ker(A - \lambda I)}$$

In simple terms:

The eigenspace contains **all eigenvectors associated with an eigenvalue, plus the zero vector**.

Notice something subtle:

The zero vector is **not an eigenvector**, but it belongs to the eigenspace.

Example

For:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

and:

$$\lambda = 3$$

we found:

$$x_1 = x_2$$

Therefore:

$$E_3 = \left\{ \begin{bmatrix} t \\ t \end{bmatrix} : t \in \mathbb{R} \right\}$$

or:

$$\boxed{E_3 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}}$$

Similarly:

$$\boxed{E_1 = \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}}$$

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12. Eigenvector vs Eigenspace

This is a common interview confusion.

Concept	Meaning
Eigenvalue	Scaling factor
Eigenvector	One non-zero vector having that scaling behavior
Eigenspace	Entire subspace associated with that eigenvalue

Example:

$$\lambda = 3$$

Eigenvectors include:

$$(1,1),\ (2,2),\ (-,7-7),\ldots$$

The eigenspace is:

$$\text{span}\{(1,1)\}$$

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13. Algebraic Multiplicity

Suppose:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Then:

$$\det(A - \lambda I) = (1-\lambda)^2(2-\lambda)$$

Therefore:

$$\lambda = 1$$

appears twice.

And:

$$\lambda = 2$$

appears once.

So:

$$\boxed{\text{Algebraic multiplicity of } 1 = 2}$$
$$\boxed{\text{Algebraic multiplicity of } 2 = 1}$$

Interview point

Algebraic multiplicity = number of times an eigenvalue appears as a root of the characteristic polynomial.

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14. Geometric Multiplicity

Geometric multiplicity is:

$$\boxed{\dim(E_\lambda)}$$

In other words:

Number of linearly independent eigenvectors associated with that eigenvalue.

For:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

for $\lambda = 1$:

$$E_1 = \{(x_1, x_2, 0)\}$$

This is a 2D plane.

Therefore:

$$\boxed{\text{Geometric multiplicity} = 2}$$

So here:

$$\text{Algebraic multiplicity} = \text{Geometric multiplicity} = 2$$

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15. The Critical Multiplicity Rule

For every eigenvalue:

$$\boxed{1 \leq GM(\lambda) \leq AM(\lambda)}$$

where:

- GM= geometric multiplicity
- AM= algebraic multiplicity

This is an important interview fact.

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16. Diagonalization

This is the major reason eigenvectors matter.

Suppose we have $n$linearly independent eigenvectors:

$$v_1, v_2, \ldots, v_n$$

Put them into a matrix:

$$P = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{bmatrix}$$

Put their eigenvalues on the diagonal:

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Then:

$$\boxed{A = PDP^{-1}}$$

or equivalently:

$$\boxed{D = P^{-1}AP}$$

This is called **diagonalization**.

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17. Why Is Diagonalization Useful?

Matrix multiplication:

$$Ax$$

can be expensive for a large dense matrix.

A diagonal matrix is much cheaper:

$$Dx = \begin{bmatrix} \lambda_1 x_1 \\ \lambda_2 x_2 \\ \vdots \\ \lambda_n x_n \end{bmatrix}$$

Only $n$multiplications are needed.

So:

$$\boxed{\text{Dense matrix multiplication: } O(n^2)}$$

versus:

$$\boxed{\text{Diagonal matrix multiplication: } O(n)}$$

This is one motivation for diagonalization.

But there is an even more important mathematical benefit:

Diagonalization exposes the independent directions in which the transformation acts simply.

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18. Diagram — Diagonalization

Think of it as changing coordinates:

Original coordinates

x

|

| A

↓

complicated

transformation

↓ change basis

Eigenvector coordinates

x

|

| D

↓

simple scaling

$\lambda_1, \lambda_2, \ldots, \lambda_n$

Instead of understanding:

“What complicated thing does A do?”

we understand:

“Along eigenvector v_1, multiply by λ_1. Along v_2, multiply by λ_2.”

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19. Example — Diagonalizing Our Matrix

We have:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Eigenvectors:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix},\ v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Eigenvalues:

$$\lambda_1 = 3,\ \lambda_2 = 1$$

Construct:

$$P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

and:

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Then:

$$\boxed{A = PDP^{-1}}$$

Therefore:

$$\boxed{P^{-1}AP = D}$$

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20. When Is a Matrix Diagonalizable?

An $n \times n$matrix is diagonalizable if it has:

$$\boxed{n \text{ linearly independent eigenvectors}}$$

This is the interview definition you should remember.

Important consequence

If a matrix has $n$distinct eigenvalues, then it is automatically diagonalizable.

Why?

Because eigenvectors corresponding to distinct eigenvalues are linearly independent.

Therefore:

$$\boxed{n \text{ distinct eigenvalues} \Rightarrow \text{diagonalizable}}$$

But the reverse isn't necessarily true.

A matrix can be diagonalizable even if some eigenvalues repeat.

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21. Example — Repeated Eigenvalue but Diagonalizable

Consider:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Eigenvalues:

$$1,1,2$$

Even though 1is repeated, we have three independent eigenvectors:

$$(1,0,0)$$
$$(0,1,0)$$
$$(0,0,1)$$

Therefore:

$$\boxed{A \text{ is diagonalizable}}$$

In fact, it is already diagonal.

---

22. Example — Not Diagonalizable

Consider:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Characteristic polynomial:

$$(1-\lambda)^2$$

So the only eigenvalue is:

$$\lambda = 1$$

Now:

$$A - I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Solve:

$$(A - I)x = 0$$

giving:

$$x_2 = 0$$

Therefore eigenvectors have form:

$$\begin{bmatrix} x_1 \\ 0 \end{bmatrix}$$

There is only **one independent eigenvector**.

But a 2×2matrix needs 2 independent eigenvectors to diagonalize.

Therefore:

$$\boxed{\text{Not diagonalizable}}$$

This is a very useful interview example.

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23. Similar Matrices

If:

$$\boxed{B = T^{-1}AT}$$

then $A$and $B$are called **similar matrices**.

They represent the same linear transformation under different coordinate systems/bases.

Important properties preserved under similarity include:

- Eigenvalues
- Characteristic polynomial
- Determinant
- Trace
- Rank

For this chapter, remember primarily:

Similar matrices have the same eigenvalues

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24. Why Does the Basis Not Change the Eigenvalues?

Suppose:

$$B = T^{-1}AT$$

Then:

$$B - \lambda I = T^{-1}AT - \lambda I$$

Since:

$$I = T^{-1}T$$

we can write:

$$B - \lambda I = T^{-1}(A - \lambda I)T$$

Taking determinant:

$$\det(B - \lambda I) = \det(T^{-1})\det(A - \lambda I)\det(T)$$

Since:

$$\det(T^{-1})\det(T) = 1$$

we get:

$$\boxed{\det(B - \lambda I) = \det(A - \lambda I)}$$

Therefore they have the same characteristic polynomial and hence the same eigenvalues.

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25. Real vs Complex Eigenvalues

Not every real matrix has real eigenvalues.

Consider:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

This represents a 90°rotation.

Its characteristic polynomial is:

$$\lambda^2 + 1 = 0$$

Therefore:

$$\lambda = \pm i$$

There are no real eigenvalues.

Geometric intuition

A pure 90°rotation changes the direction of every non-zero real vector.

So there is no real direction that remains on the same line.

This is an excellent interview intuition.

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26. Implementation — Python / NumPy

For ML interviews, you should know how to calculate eigenvalues/eigenvectors computationally.

```
import numpy as np


A = np.array([
   [2, 1],
   [1, 2]
])


eigenvalues, eigenvectors = np.linalg.eig(A)


print("Eigenvalues:")
print(eigenvalues)
```

print("Eigenvectors:")

print(eigenvectors)

Typical result:

Eigenvalues:

[3. 1.]

Eigenvectors:

[[-0.70710678 0.70710678]

 [-0.70710678 -0.70710678]]

The exact signs/order may differ.

For example:

$$\begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

is equivalent to:

$$\begin{bmatrix} -0.707 \\ -0.707 \end{bmatrix}$$

because eigenvectors can be multiplied by any non-zero scalar.

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27. Verify the Result in Code

Never blindly trust a numerical result.

For each eigenpair:

$$Av \approx \lambda v$$

You can verify:

```
for i in range(len(eigenvalues)):
    lam = eigenvalues[i]
    v = eigenvectors[:, i]
```

```
print(np.allclose(A @ v, lam * v))
```

Output:

True

True

Interview mindset

This is important for ML engineering:

Compute → verify → interpret

rather than:

Compute → assume it's correct.

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28. Finding Eigenvalues Without NumPy

For a 2×2matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Characteristic polynomial:

$$\det(A - \lambda I) = 0$$

gives:

$$\begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = 0$$

Therefore:

$$(a-\lambda)(d-\lambda) - bc = 0$$

Expanding:

$$\boxed{\lambda^2 - (a+d)\lambda + (ad - bc) = 0}$$

Notice:

$$a + d = \mathrm{tr}(A)$$

and:

$$ad - bc = \det(A)$$

Therefore:

$$\boxed{\lambda^2 - \operatorname{tr}(A)\lambda + \det(A) = 0}$$

This is a very useful 2×2 interview shortcut.

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29. Trace and Eigenvalues

For an $n \times n$matrix:

$$\boxed{\operatorname{tr}(A) = \sum_i \lambda_i}$$

And:

$$\boxed{\det(A) = \prod_i \lambda_i}$$

counting algebraic multiplicity.

For our matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Trace:

$$2 + 2 = 4$$

Eigenvalues:

$$3 + 1 = 4$$

Correct.

Determinant:

$$4 - 1 = 3$$

Product:

$$3 \times 1 = 3$$

Correct.

Interview trap

If someone asks:

"What is the relationship between eigenvalues and determinant?"

Answer:

$$\boxed{\det(A) = \prod_i \lambda_i}$$

not the sum.

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30. Singular Matrix and Eigenvalue 0

A matrix is singular when:

$$\det(A) = 0$$

Since:

$$\det(A) = \prod_i \lambda_i$$

we have:

$$\boxed{A \text{ singular } \iff 0 \text{ is an eigenvalue}}$$

This is another high-value interview connection.

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31. Powers of a Matrix

Diagonalization becomes particularly useful when computing:

$$A^k$$

If:

$$A = PDP^{-1}$$

then:

$$A^2 = PDP^{-1}PDP^{-1}$$

Since:

$$P^{-1}P = I$$

we get:

$$A^2 = PD^2P^{-1}$$

Similarly:

$$\boxed{A^k = PD^kP^{-1}}$$

And:

$$D^k = \begin{bmatrix} \lambda_1^k & 0 & \cdots \\ 0 & \lambda_2^k & \cdots \\ & & \ddots \end{bmatrix}$$

This is much easier than multiplying $A$by itself repeatedly.

---

32. Why Eigenvalues Matter in Machine Learning

This is where the chapter becomes relevant to ML.

Eigenvalues/eigenvectors appear heavily in:

1. PCA

Principal Component Analysis uses eigenvectors of the covariance matrix.

The eigenvectors identify important directions in the data.

The corresponding eigenvalues indicate how much variance lies along those directions.

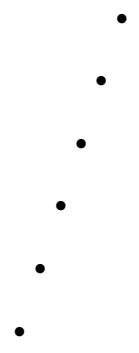
Very important:

Large eigenvalue ⇒ direction with large variance

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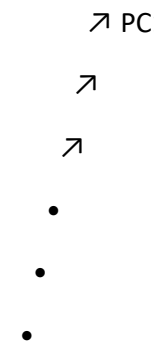
33. PCA Diagram

Imagine a dataset shaped like this:



The data has a dominant direction.

PCA finds approximately:



The first principal component is the eigenvector associated with the **largest eigenvalue** of the covariance matrix.

The second principal component is another orthogonal eigenvector associated with the next-largest eigenvalue.

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34. Why Eigenvalues Matter for PCA

Suppose covariance matrix eigenvalues are:

$$10,\ 3,\ 1$$

Total variance:

Explained variance ratio of first component:

$$\frac{10}{14} \approx 71.4\%$$

Second:

$$\frac{3}{14} \approx 21.4\%$$

Third:

$$\frac{1}{14} \approx 7.1\%$$

So the first two components explain:

$$\frac{13}{14} \approx 92.9\%$$

This lets us reduce 3 dimensions to 2 while retaining most of the variance.

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35. Eigenvalues and Stability

Eigenvalues also appear in dynamical systems.

Suppose:

$$x_{t+1} = Ax_t$$

If Ahas eigenvalues:

$$\lambda_1, \lambda_2, \ldots$$

then repeated application gives behavior related to:

$$\lambda_i^t$$

Therefore:

- $|\lambda| < 1 \rightarrow$ tends to decay

- $|\lambda| > 1 \rightarrow$ tends to grow
- $|\lambda| = 1 \rightarrow$ can persist

This idea appears in:

- optimization
- numerical methods
- recurrent systems
- dynamical systems
- stability analysis

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36. Eigenvalues and Neural Networks — Interview Level

In deep learning, eigenvalues can help analyze:

- Hessian matrices
- curvature
- optimization
- conditioning
- gradient behavior

For example, if the Hessian has very different eigenvalues:

$$\lambda_{\max} \gg \lambda_{\min}$$

the optimization landscape can be poorly conditioned.

This can cause optimization to move much faster in some directions than others.

For your ML-engineering path, understand this **conceptually first** rather than diving immediately into advanced Hessian theory.

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37. Eigenvalues vs Singular Values

This is an important interview distinction.

Eigenvalue

Defined by:

$$Av = \lambda v$$

It asks:

Does Amap this vector to a scalar multiple of itself?

Singular value

Obtained from:

$$A^T A$$

and is related to:

$$\sigma_i = \sqrt{\lambda_i(A^T A)}$$

Singular values measure scaling of orthogonal directions even when ordinary eigenvectors aren't sufficient.

Interview distinction

Eigenvalues	**Singular values**
$Av = \lambda v$	Related to $A^T A$
Can be complex	Non-negative real
Mainly square matrices	Works for rectangular matrices
Special invariant directions	Principal stretching factors
PCA/eigensystems	SVD, PCA, dimensionality reduction

Don't confuse these.

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38. The Most Important Formulas

Create this mental cheat sheet:

Definition

$$\boxed{Av = \lambda v}$$

Characteristic equation

$$\boxed{\det(A - \lambda I) = 0}$$

Eigenvectors

$$\boxed{(A - \lambda I)v = 0}$$

Eigenspace

$$\boxed{E_\lambda = \ker(A - \lambda I)}$$

Diagonalization

$$\boxed{A = PDP^{-1}}$$

Equivalent form

$$\boxed{D = P^{-1}AP}$$

Trace

$$\boxed{\operatorname{tr}(A) = \sum_i \lambda_i}$$

Determinant

$$\boxed{\det(A) = \prod_i \lambda_i}$$

Matrix powers

$$\boxed{A^k = PD^kP^{-1}}$$

Diagonalizability

$$\boxed{A \text{ diagonalizable } \iff A \text{ has } n \text{ independent eigenvectors}}$$

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39. 80/20 Interview Priority

Don't spend equal time on every section.

Concept	Priority	What you should be able to do
$Av = \lambda v$	🔴 P0	Explain intuitively
Eigenvector/eigenvalue meaning	🔴 P0	Explain geometrically
Characteristic equation	🔴 P0	Derive it
Finding eigenvalues	🔴 P0	Solve 2×2/3×3 problems
Finding eigenvectors	🔴 P0	Solve $(A - \lambda I)v = 0$
Eigenspace	🔴 P0	Explain and calculate
Diagonalization	🔴 P0	Explain $A = PDP^{-1}$
Distinct eigenvalues	🔴 P0	Explain why diagonalizable
Trace/determinant relationship	🔴 P0	Use as shortcut/check
Algebraic multiplicity	🟠 P1	Understand
Geometric multiplicity	🟠 P1	Understand
Similar matrices	🟠 P1	Explain basis invariance
Complex eigenvalues	🟠 P1	Explain rotation example
NumPy implementation	🔴 P0	Use np.linalg.eig
PCA connection	🔴 P0 for ML	Explain eigenvectors/eigenvalues
Hessian/optimization	🟠 P1	Understand conceptually
Advanced spectral theory	🟢 P2	Defer

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40. Interview Problem-Solving Template

When given:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

follow this exact process.

Step 1 — Write the definition

$$Av = \lambda v$$

Step 2 — Rearrange

$$(A - \lambda I)v = 0$$

Step 3 — Require non-zero solution

$$\det(A - \lambda I) = 0$$

Step 4 — Solve for λ

These are the eigenvalues.

Step 5 — For each eigenvalue

Solve:

$$(A - \lambda I)v = 0$$

Step 6 — Get eigenvectors

Any non-zero solution is an eigenvector.

Step 7 — Find eigenspace

Write:

$$E_\lambda = \text{span}\{v_1, v_2, \dots\}$$

Step 8 — Check diagonalizability

Do you have $n$linearly independent eigenvectors?

If yes:

$$A = PDP^{-1}$$

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41. A Complete Interview Example

Given:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Step 1: Characteristic polynomial

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

Therefore:

$$\det(A - \lambda I) = (4-\lambda)(3-\lambda) - 2$$

Expand:

$$12 - 7\lambda + \lambda^2 - 2$$
$$= \lambda^2 - 7\lambda + 10$$

Factor:

$$(\lambda - 5)(\lambda - 2) = 0$$

Therefore:

$$\boxed{\lambda_1 = 5,\ \lambda_2 = 2}$$

Step 2: Eigenvector for $\lambda = 5$

$$A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

Solve:

$$-x + y = 0$$

so:

$$x = y$$

Choose:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Step 3: Eigenvector for $\lambda = 2$

$$A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

Solve:

$$2x + y = 0$$

so:

$$y = -2x$$

Choose:

$$v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Step 4: Diagonalization

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$$

and:

$$D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

Therefore:

$$\boxed{A = PDP^{-1}}$$

This is exactly the kind of problem you should be able to solve on a whiteboard.

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42. Common Interview Traps

Trap 1: Zero vector is an eigenvector

❌ Wrong.

Eigenvector must satisfy:

$$\boxed{v \neq 0}$$

Otherwise every λwould satisfy:

$$A0 = \lambda 0$$

which would make the definition meaningless.

Trap 2: Eigenvector remains unchanged

❌ Not necessarily.

It can be:

$$Av = 3v$$

or:

$$Av = -2v$$

The direction/line remains invariant.

Trap 3: Every matrix is diagonalizable

❌ Wrong.

It requires enough linearly independent eigenvectors.

Trap 4: Every real matrix has real eigenvalues

❌ Wrong.

The rotation matrix:

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

has eigenvalues:

$$\pm i$$

Trap 5: Repeated eigenvalue means not diagonalizable

❌ Wrong.

Repeated eigenvalues can still have enough independent eigenvectors.

Trap 6: Distinct eigenvalues guarantee distinct eigenvectors

More precisely:

> Eigenvectors corresponding to distinct eigenvalues are linearly independent

This is why distinct eigenvalues guarantee diagonalizability for an $n \times n$matrix.

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43. Interview Questions You Should Master

Basic

1. What is an eigenvalue?
2. What is an eigenvector?
3. Explain $Av = \lambda v$geometrically.
4. Why can't the zero vector be an eigenvector?
5. What does a negative eigenvalue mean?

Calculation

6. How do you find eigenvalues?
7. Why do we solve $\det(A - \lambda I) = 0$?
8. How do you find eigenvectors after finding eigenvalues?
9. Find eigenvalues/eigenvectors of a 2×2matrix.
10. Find the eigenspace.

Intermediate

11. What is algebraic multiplicity?
12. What is geometric multiplicity?
13. What is diagonalization?
14. When is a matrix diagonalizable?
15. Why do distinct eigenvalues imply diagonalizability?
16. What happens if an eigenvalue is repeated?
17. What are similar matrices?

18. Why are eigenvalues invariant under change of basis?

ML

19. How are eigenvalues used in PCA?
20. Why does PCA choose eigenvectors of the covariance matrix?
21. What does a large eigenvalue mean in PCA?
22. What is the relationship between eigenvalues and variance?
23. Eigenvalues vs singular values?
24. Why do eigenvalues matter in optimization?

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44. The 60-Second Interview Explanation

If the interviewer says:

"Explain eigenvalues and eigenvectors."

A strong answer is:

"For a square matrix A, an eigenvector is a non-zero vector $v$whose direction is preserved by the transformation. Mathematically, $Av = \lambda v$, where $\lambda$is the eigenvalue and represents the scaling factor. To find eigenvalues, we rearrange this as $(A - \lambda I)v = 0$. For a non-zero solution to exist, $A - \lambda I$must be singular, so we solve $\det(A - \lambda I) = 0$. After finding each eigenvalue, we solve $(A - \lambda I)v = 0$to find its eigenvectors and eigenspace. If an $n \times n$matrix has $n$linearly independent eigenvectors, it can be diagonalized as $A = PDP^{-1}$. In machine learning, eigenvalues and eigenvectors are particularly important in PCA, where eigenvectors give principal directions and eigenvalues indicate the variance along those directions."

That is a **very strong 80/20 answer**.

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45. Final Mental Model

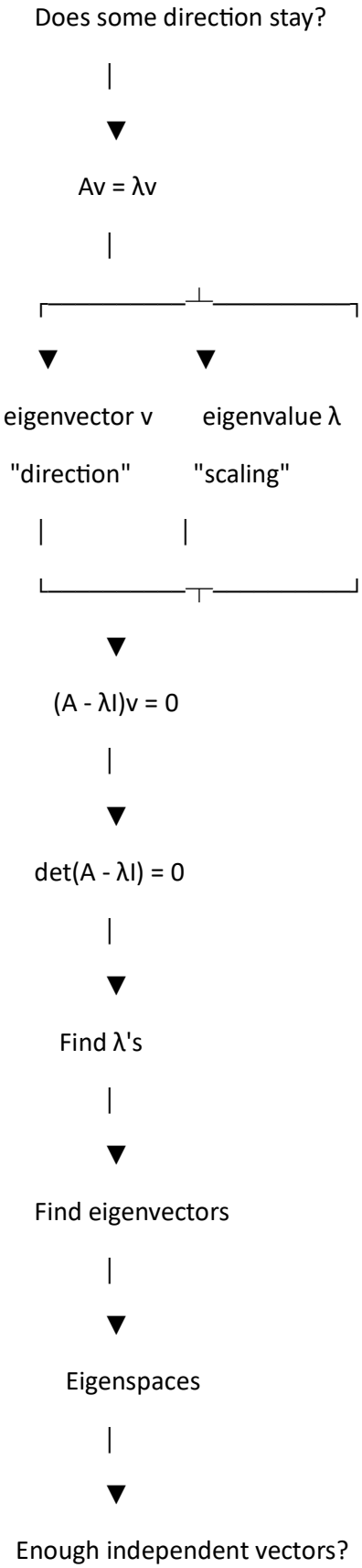
Don't memorize this chapter as 20 disconnected definitions.

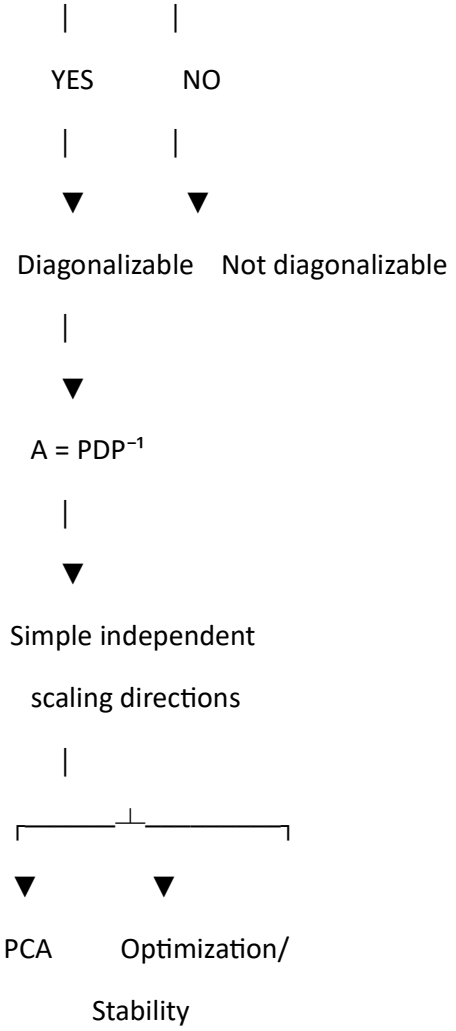
Remember this one flow:

MATRIX A

|

▼


Does some direction stay?
Av = λv
eigenvector v
eigenvalue λ
"direction"
"scaling"
(A - λI)v = 0
det(A - λI) = 0
Find λ's
Find eigenvectors
Eigenspaces
Enough independent vectors?



What to actually practice

For your next interview, prioritize these **5 capabilities**:

1. **Explain $Av = \lambda v$geometrically without memorization.**
2. **Calculate eigenvalues manually for 2×2 and simple 3×3 matrices.**
3. **Calculate eigenvectors by solving $(A - \lambda I)v = 0$.**
4. **Determine whether a matrix is diagonalizable and explain why.**
5. **Connect eigenvalues/eigenvectors to PCA and explain why large eigenvalues correspond to important variance directions.**

If you can do those five confidently, you have captured most of the **interview-relevant value of this chapter**.